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FACULTY OF INFORMATION AND COMMUNICATION TECHNOLOGY

DATA STRUCTURES AND ALGORITHM DCOMP320

Title : INDIVIDUAL ASSIGNMENT
Issue Date : Week 2
Due Date : Week 4
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Class : BIT 2101
Semester/Year : 5

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STACKS IMPLEMENTATION

INDIVIDUAL ASSIGNMENT

DATA STRUCTURE & ALGORITHM DCOMP320

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7 OCTOBER 2026

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1. INTRODUCTION

Stacks operation helps evaluate mathematical expressions by following the Last-In, First-Out (LIFO) principle. An operand stack stores numbers, while an operator stack stores mathematical operators. The operator stack helps the program follow PEMDAS/BODMAS by processing multiplication and division before addition and subtraction. Parentheses are also managed by pushing opening brackets onto the stack and evaluating the operators inside them when a closing bracket is found. For example, in 10 - (2 + 3) * 4, the brackets are evaluated first, followed by multiplication and then subtraction, giving **-** 10. Thus, stacks ensure that mathematical operations are performed in the correct order.

The objective of this assignment is to develop a stack class with push, pop, and peek operations, parse and evaluate mathematical expressions using the operators **+, -, *,** and **/**, apply parentheses and operator precedence rules, and correctly manage input and output through files.

The program reads a text file (input.txt) containing several expressions separated by lines of dashes, evaluates each expression, and writes the results to a second text file (output.txt) using the same layout and separators.

2. DATA STRUCTURES USED

2.1 The Stack Data Structure

A stack represents a linear data structure governed by Last In – First Out (LIFO) methodology, restricting data insertion and deletion operations strictly to the terminal index (the "top"). A stack supports the following core operations:

- push: adds an item to the top of the stack.
- pop: removes and returns the item on top of the stack.
- peek: returns the item on top without removing it.

Each of these operations takes constant time, O(1), because they only affect one end of the structure. In this program the stack is implemented as a class that wraps a Python list. Popping or peeking at an empty stack raises an error, which the evaluator reports as an invalid expression rather than crashing.

2.2 How the Stack Is Used to Evaluate Expressions

The evaluator uses two stacks, both implemented using the same Stack class. One stack stores numerical operand, while the other stores operators and opening parentheses. The expression is first split into tokens (numbers, operators and parentheses) and then processed from left to right according to the following rules:

- A number is pushed onto the operand stack.
- An opening parenthesis is pushed onto the operator stack.

A closing parenthesis causes operators to be popped and applied until the matching opening parenthesis is reached, which is then discarded.

- When an operator is read, any operator already on the stack with higher or equal precedence is applied first. The new operator is then pushed. This produces the correct precedence and left-to-right order.
- At the end of the expression, all remaining operators are applied. The single value left on the operand stack is the result.

Applying an operator means popping the required operands, performing the calculation, and pushing the result back onto the operand stack. For example, 3 + 5 * 2 is evaluated as follows:

Token	Operand Stack	Operator Stack	Action
3	[3]	[ ]	Push operand
+	[3]	[+]	Push operator
5	[3, 5]	[+]	Push operand
*	[3, 5]	[+, *]	* outranks +, so push
2	[3, 5, 2]	[+, *]	Push operand
End	[3, 10]	[+]	Apply *: 5 * 2 = 10
End	[13]	[ ]	Apply +: 3 + 10 = 13

2.3 Importance of Stacks in Computing

Stacks are fundamental to computing because they manage execution flow, memory allocation, and data parsing through the Last-In, First-Out (LIFO) protocol. Without stacks, modern software architectures would struggle to handle basic functions like running applications, navigating the web, or compiling code. Compilers and interpreters rely on stacks to analyse program syntax, verify balanced brackets, and evaluate or convert expressions.

Similarly, text editors and other applications use stacks to support undo functions, while web browsers use them to manage and navigate back through browsing history. Depth-first search (DFS) in graph algorithms also makes use of stacks to explore nodes systematically. Since stack operations are simple and efficient, stacks are well suited for situations where the most recently added item needs to be processed first, such as when handling nested parentheses in mathematical expressions. Text editors and other software applications use stacks to implement undo operations, while web browsers rely on stacks to store and manage users' browsing history when navigating backward.

3. PROGRAM DESIGN

3.1 Program Architecture

The program is contained in a single Python file, main.py, and is organised into three layers. The data structure layer contains the Stack class. The processing layer contains the tokenizer and the evaluator, which together convert an expression string into a numerical result. The file handling layer reads the input file, calls the evaluator for each expression, and writes the formatted results to the output file. The overall flow is read input.txt, split into expressions at each separator line, tokenize and evaluate each expression, format the result, and write output.txt.

3.2 Key Functions and Classes

Function / Class	Description
Stack	Implements push, pop, peek, is_empty and size using a Python list. Raises an ExpressionError if pop or peek is used on an empty stack.
tokenize(expression)	Splits the expression into numbers, operators and parentheses using a regular expression. Converts typographic symbols such as the dash (–) to a standard minus sign and rejects invalid characters.
apply_operator(operators, operands)	Pops one operator and the needed operand(s), performs the calculation, and pushes the result. Raises an error for division by zero.
evaluate(expression)	Core algorithm. Processes tokens using the operand and operator stacks, applying precedence rules, parentheses and unary minus, and validates the expression structure.
format_result(value)	Displays whole numbers without a decimal point (13 instead of 13.0) and rounds other values to six decimal places.
read_blocks(path)	Reads the input file and returns the list of expressions, using lines made only of dashes or similar characters as separators. Blank lines are ignored.
process_file(input, output)	Evaluates every expression and writes the results to the output file, reproducing the separators. Invalid expressions produce an error message line instead of stopping the program.

4. TESTING AND RESULTS

4.1 Testing Approach

The program was tested in two stages. The three expressions given in the assignment were placed in input.txt and the program was run to generate output.txt. Each result was compared with the value calculated by hand.

4.2 Results

No.	Expression	Program output	Test purpose
1	3 + 5 * 2	13	Precedence
2	(8 / 4) + 7 * 2	16	Parentheses and precedence
3	10 – (2 + 3) * 4	-10	dash and parentheses
4	2 * (3 + 4) * 5	70	Multiple operators
5	100 / 8 / 2	6.25	Left-to-right division

4.3 Discussion

For the input.txt file, the program produced 13, 16 and -10, with the same separator lines placed between the results as in the input.

One point should be noted. The example output in the assignment brief lists 17 as the result of (8 / 4) + 7 * 2. Calculating by hand, 8 / 4 = 2 and 7 * 2 = 14, so the correct answer is 2 + 14 = 16. The program's result of 16 is therefore correct, and the value 17 in the brief appears to be a typing error.

The program has some limitations. It supports only the operators +, -, * and / and does not handle exponents, functions or variables. Division uses floating-point arithmetic, so some results are rounded to six decimal places.

5. CONCLUSION

This assignment demonstrates a robust Python-based expression evaluator utilizing a standardized Stack architecture. By employing dual-stack processing logic, the system successfully resolves token precedence, parenthesis grouping, and unary operators for datasets extracted from input.txt. Output formatting constraints were strictly maintained within output.txt, with rigorous testing verifying resilient parsing of malformed inputs. The project highlights the computational efficiency of LIFO structures, proving that elementary data abstractions can eliminate the need for over-engineered control flows in language processing.

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